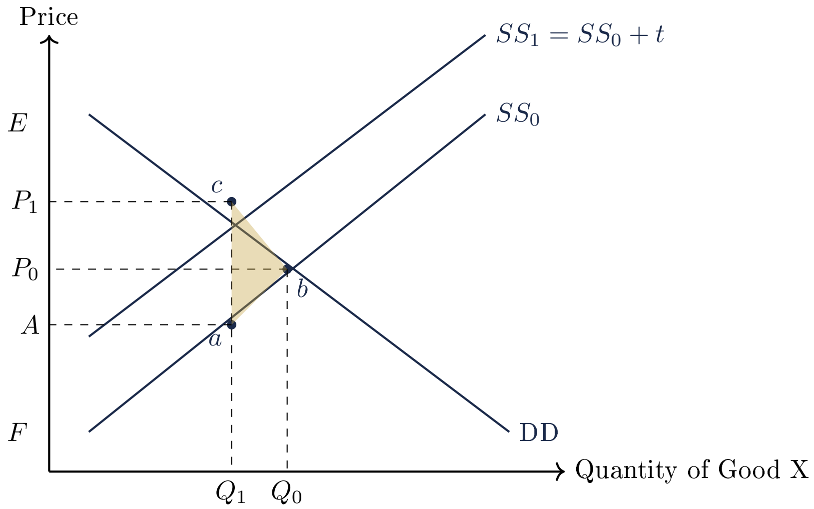
Document 10. Government Intervention in Markets

Topic overview. Governments intervene in markets to resolve market failure — the failure of the free market to achieve allocative efficiency or equity. Four instruments are commonly used: indirect taxes, indirect subsidies, price controls, and quantity controls. This cheat sheet provides the Crucible verbatim frameworks for each.

Section A. Important Definitions

Specific tax	Constant sum levied on each unit of the good sold. Parallel leftward shift of SS.
Ad valorem tax	Tax pegged at a percentage of the good. Leftward pivotal shift of SS.
Indirect subsidy	Transfer from government to producer per unit produced. Parallel rightward shift of SS.
Price floor	Legally established minimum price set above market equilibrium.
Price ceiling	Legally established maximum price set below market equilibrium.
Quota	Limit on the quantity produced imposed by government, set below market equilibrium quantity.
Consumer Surplus (CS)	Difference between the highest price consumers are willing to pay and the price they actually pay.
Producer Surplus (PS)	Difference between the lowest price producers are willing to supply at and the price they actually receive.
Deadweight Loss (DWL)	Measure of the loss of economic efficiency when allocative efficiency is not achieved.
Tax incidence	Distribution of the burden of taxation between consumers and sellers. PED $<$ PES $\Rightarrow$ consumers bear more; PED $>$ PES $\Rightarrow$ producers bear more.

Section B. Diagram — Specific Tax (with welfare areas)



Welfare summary: Tax revenue $= P_1caA$. New CS $= P_1cE$. New PS $= AaF$. DWL $= cba$.

Section C. Framework: Specific Tax when PED $<$ PES (verbatim Crucible)

Step 1. In the market for *(Good X)*, producers are more responsive to price changes ($PES > 1$) and consumers are less responsive to price changes ($PED < 1$).
Step 2. The price elasticity of supply is greater than one because *(insert factor)*; while the price elasticity of demand is less than one because *(insert factor)*.
Step 3. Indirect specific taxes are payments charged to producers of the good by the government per unit of good produced. For example, the Singapore government levies taxes on each pack of cigarettes.
Step 4. The imposition of a specific indirect tax causes a parallel leftward shift of the supply curve as it increases the costs of production for producers. Ceteris paribus, supply falls because producers are disincentivised by the lower potential profit per unit.
Step 5. The equilibrium price increases by P_0–P_1. However, the increase in price per unit for consumers is less than the tax per unit ($ab < ac$). The producers bear a lower per unit tax incidence of bc, while consumers bear a higher per unit tax incidence of ac.

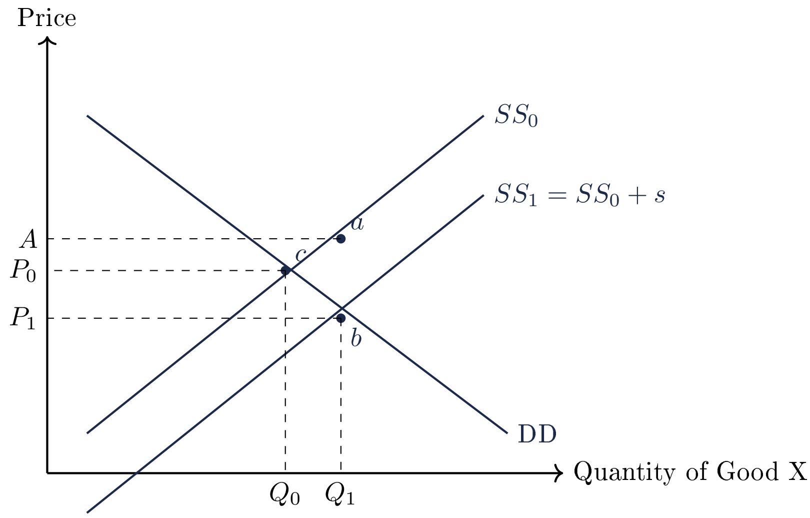
Section D. Framework: Specific Tax when PES < PED (verbatim Crucible)

Step 1. In the market for *(Good X)*, consumers are highly responsive to price changes ($PED > 1$) and producers are less responsive to price changes ($PES < 1$).
Step 2. The price elasticity of supply is less than one because *(insert factor)*; while the price elasticity of demand is greater than one because *(insert factor)*.
Step 3. Indirect specific taxes are payments charged to producers of the good by the government per unit of good produced.
Step 4. The imposition of a specific indirect tax causes a parallel leftward shift of the supply curve as it increases the costs of production for producers.
Step 5. The equilibrium price increases by P_0P_1. The producers bear a higher per unit tax incidence of bc and consumers bear a lower per unit tax incidence of ab.

Section E. Impact of Indirect Taxes on Stakeholders

- **Consumers** are *worse off*: pay higher prices by P_0P_1, enjoy lower quantities by Q_0Q_1. CS reduced by P_1cbP_0.
- **Producers** (if PED < 1) are *better off*: higher price causes less than proportionate fall in Q_d. TR rises; PS rises by $P_0bF - AaF$. (If PED > 1, the reverse holds.)
- **Government** is *better off*: receives tax revenue of P_1caA. Positive for the Budget; revenue can be redirected to maximise societal welfare.
- **Workers** may be *worse off*: lower output by Q_0Q_1 means fewer workers required (labour is derived demand). Unemployment may increase.
- **Society** is *worse off*: under-allocation of resources to (Good X) by Q_0Q_1. Allocative inefficiency results.

Section F. Diagram — Per-Unit Subsidy



Welfare summary: Government expenditure = $AabP_1$. New CS = P_1Eb. New PS = AaF. DWL = cba.

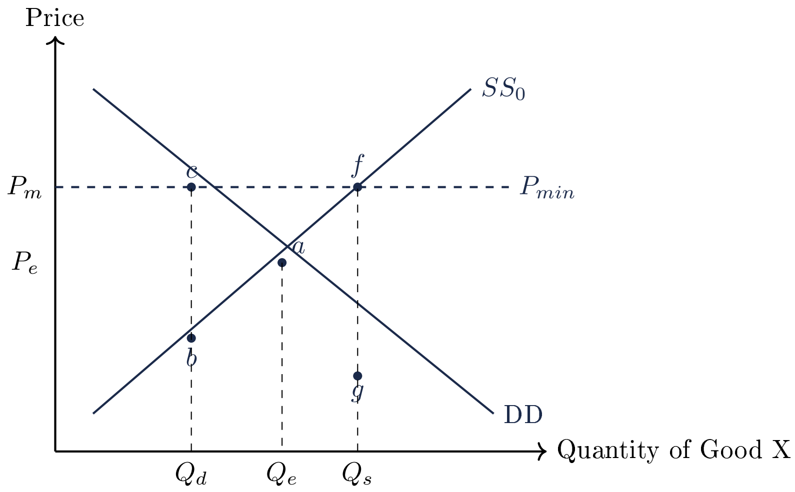
Section G. Framework: Subsidy when PED < PES (verbatim Crucible)

Step 1. In the market for *(Good X)*, producers are highly responsive to price changes ($PES > 1$) and consumers are less responsive to price changes ($PED < 1$).
Step 2. The price elasticity of supply is greater than one because *(insert factor)*; while the price elasticity of demand is less than one because *(insert factor)*.
Step 3. Indirect specific subsidies are transfers from the government to the producer for the provision of goods that are given per unit of the good produced. For example, the Singapore government subsidises each visit to the polyclinic for all citizens.
Step 4. The imposition of a specific indirect subsidy causes a parallel rightward shift of the supply curve as it decreases the costs of production for producers. Ceteris paribus, supply increases because producers are incentivised by the higher potential profit per unit.
Step 5. The equilibrium price decreases by P_0P_1. However, the decrease in price per unit for consumers is less than the subsidy per unit ($ab < ac$). The producers enjoy a lower per unit subsidy incidence of bc and consumers enjoy a higher per unit subsidy incidence of ab.

Section H. Impact of Subsidies on Stakeholders

- **Consumers** are *better off*: pay lower prices by P_0P_1, enjoy increased quantities by Q_0Q_1. CS increased by P_1bcP_0.
- **Producers** (if PED > 1) are *better off*: lower price causes more than proportionate rise in Q_d; TR rises. (If PED < 1, the reverse holds.)
- **Government** is *worse off*: incurs subsidy expenditure of $AabP_1$. Negative for Budget; opportunity cost.
- **Workers** may be *better off*: higher output requires more labour.
- **Society** is *worse off*: over-allocation of resources by Q_0Q_1. Allocative inefficiency.
- **Foreign producers** may be *worse off* if (Good X) is exported: lowered domestic price erodes their competitiveness.

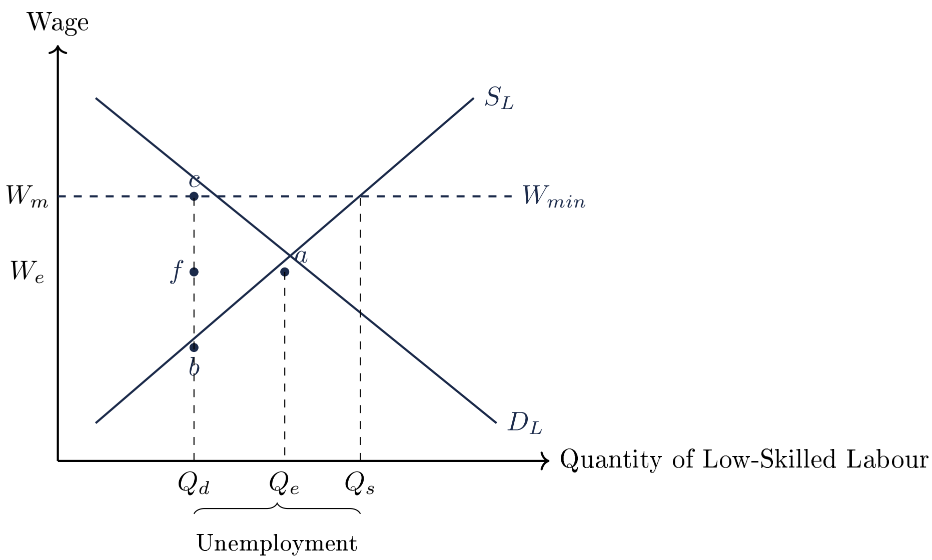
Section I. Diagram — Price Floor on Product Markets



Section J. Framework: Price Floor on Product Markets (verbatim Crucible)

Step 1. A price floor is a legally established minimum price set above the market equilibrium price to prevent prices from falling below a certain level. For example, the US government uses price floors to protect the income of American corn farmers.
Step 2. The free market equilibrium price and quantity is at P_e and Q_e respectively. Before the implementation of the price floor, the total revenue is $0P_eaQ_e$.
Step 3. After the price floor is implemented above the market equilibrium at P_m, quantity demanded falls to Q_d and quantity supplied rises to Q_s.
Step 4. Incentivised by the higher prices, firms increase the factors allocated to the production of the good, and quantity supplied increases to Q_s. However, at P_m, consumers are only prepared to buy output up to Q_d, therefore there is a surplus of Q_dQ_s.
Step 5. The surplus created of Q_dQ_s remains because the market is prevented from clearing, and the government is usually forced to buy up the surpluses from producers for storage or export. Allocative efficiency is also worsened because too much scarce resources are allocated to the production of the good.

Section K0. Diagram — Wage Floor on Labour Market



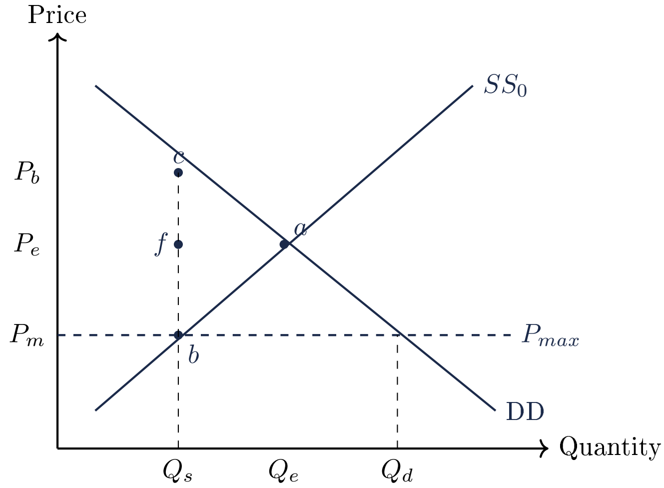
Section K. Framework: Price Floor on Labour Markets (Wage Floor, verbatim Crucible)

Step 1. A wage floor is a legally established minimum price set above the market equilibrium price to prevent wages from falling below a certain level. For example, the US government uses wage floors to protect low-income workers.
Step 2. The free market equilibrium price and quantity is at W_e and Q_e respectively.
Step 3. After the wage floor is implemented above the market equilibrium at W_m, quantity demanded falls to Q_d and quantity supplied rises to Q_s.
Step 4. Incentivised by the higher wages, more workers seek work, and quantity supplied increases to Q_s. However, at W_m, firms are only prepared to hire workers up to Q_d, therefore there is a surplus of labour by Q_dQ_s.
Step 5. The surplus created of Q_dQ_s remains because the market is prevented from clearing, representing the level of unemployment created by the wage floor. Allocative efficiency is also worsened because too much scarce resources are allocated to producing low-skilled labour.

Section L. Unintended Consequences of Wage Floors

- **Increased unemployment.** Surplus of labour Q_dQ_s; unskilled workers are most affected.
- **Increased labour costs and worsened future employment.** Firms switch to labour-replacing technology in the long-run.
- **Black markets.** Illegal employment below the minimum wage, often involving illegal immigrants.
- **Misallocation in product markets.** Higher labour costs raise product prices and reduce supply.

Section M. Diagram — Price Ceiling



Section N. Framework: Price Ceiling (verbatim Crucible)

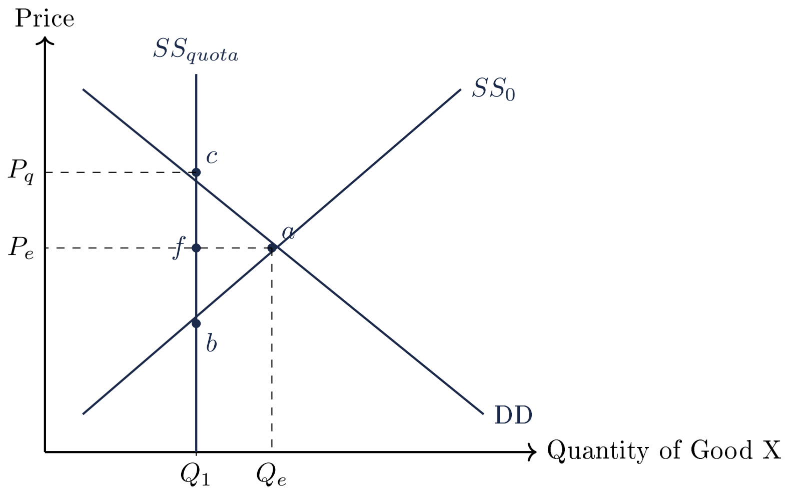
Step 1. A price ceiling is a legally established maximum price set below the market equilibrium price to prevent prices from rising above a certain level. For example, Kenya has used price ceilings on maize to keep it affordable for the locals.
Step 2. The free market equilibrium price and quantity is at P_e and Q_e respectively. Before the

implementation of the price ceiling, the total revenue is $0P_eaQ_e$.
Step 3. After the price ceiling is implemented below the market equilibrium at P_m, quantity demanded rises to Q_d and quantity supplied falls to Q_s.
Step 4. Disincentivised by the lower prices, firms decrease the factors allocated to the production of the good, and quantity supplied decreases to Q_s. However, at P_m, consumers are willing to buy an increased output up to Q_d. Therefore, there is a shortage of Q_dQ_s.
Step 5. The shortage created of Q_dQ_s remains because the market is prevented from clearing. Allocative efficiency is also worsened because too few scarce resources are allocated to the production of the good.

Section O. Unintended Consequences of Price Ceilings

- **Shortages.** Some consumers cannot access the good; lower-income groups especially affected.
- **Non-price rationing.** Queues, coupons, preferential selling — may worsen equity.
- **Black markets.** Goods sold illegally above P_m at price P_b, gaining additional revenue P_mbcP_b.
- **Increased opportunity costs for government.** May need to release stocks, subsidise producers, or directly produce.
- **Allocative inefficiency.** Too few resources allocated to (Good X).

Section P0. Diagram — Quota



Welfare summary: New CS $= P_qcE$. New PS $= P_qcbf$. DWL $= cba$. Quota rent (to licence-holders) $= P_qcfP_e$.

Section P. Framework: Quota (verbatim Crucible)

Step 1. The initial market equilibrium price and quantity is at P_e and Q_e respectively.
Step 2. A quota is a limit on the quantity produced imposed by a government through legislation that is set below the market equilibrium quantity. For example, the COE scheme in Singapore limits the number of permitted new car ownerships each month.
Step 3. After the quota has been implemented, supply shifts leftwards and becomes perfectly inelastic at Q_1. It now becomes illegal to produce beyond Q_1.
Step 4. Consequently, the market equilibrium output decreases to Q_1 and the market equilibrium

price increases to P_q.
Step 5. Thus, the price mechanism is displaced, and the output is no longer determined by the free market.

Section Q. Singapore Applications

- **GST** (ad valorem 9%); **excise duties** on tobacco, alcohol, petrol.
- **Carbon tax**: S$25/t$CO_2$e (2024), rising to S$45 by 2026–27.
- **COE quota system**: cap on new cars; competitive bidding.
- **Progressive Wage Model** in cleaning, security, landscaping, retail, food services, waste.
- **Subsidies**: U-Save, GST Vouchers, Workfare, MediShield Life premium subsidies, polyclinic visits, education.

NOTE: All five frameworks (specific tax $\times$ 2 elasticity cases, subsidy, price floor product, wage floor, price ceiling, quota) reproduced word-for-word from the Crucible Education Centre handout.